Contents

Project: Azure Capacity Reservation Management Engine (ACRME)
Classification: Principal Cloud Architect — Architecture Governance
Version: 1.1
Date: August 2026
Status: Accepted
Part of: ACRME Architecture Decision Records — this is one of four standalone ADRs split from the consolidated ADR set.

> **About ADRs.** An Architecture Decision Record captures a single significant architectural decision, the context that forced it, the options considered, the choice made, and its consequences. ADRs are immutable once accepted — a superseding decision is recorded as a new ADR rather than editing the original. Evidence tags: `[Documented]`, `[Decided]`, `[Derived]`, `[Assumed]` (see Appendix).

ADR-002 — Quota and Capacity Management

Status: Accepted
Date: August 2026
Deciders: Principal Cloud Architect, Platform Engineering, FinOps
Related decisions: D6, D7, D9; HC-2, HC-3, HC-7
Source: `multi_region_placement_design.md` §Quota Groups; `design_change_summary.md` QG-1..QG-12

Context

Each customer needs three CRGs per region (Prod, NonProd, DR). Azure tracks compute quota per subscription per region per SKU family; quota exhaustion silently blocks CR creation and VM deployment. The engine must:

- Prevent a NonProd surge from consuming quota that Prod CR creation needs (isolation).
- Make emergency capacity transfer (ADR-003) **quota-neutral** where possible — i.e. reshuffle capacity without waiting hours for an Azure quota-increase approval during a crisis.
- Provide clean per-environment cost attribution for chargeback.
- Enforce a protected DR reservation that NonProd growth can never erode.

Azure **Quota Groups** (Preview) allow multiple subscriptions/CRGs to share a group-level quota pool, but Azure does **not** natively support intra-group sub-reservations — so any sub-partition of a group pool must be enforced by the engine.

Decision

Adopt a **Two-Quota-Group-per-region model with an engine-enforced DR floor**:

1. **Two groups per region:** one **Prod-only** group and one **shared NonProd+DR** group. `[Decided - D6]`

   ```
   Prod_Group_Limit(R)        = base_subscription_quota_limit × (1 + emergency_transfer_headroo
   NonProd_DR_Group_Limit(R)  = base_subscription_quota_limit × (1 + emergency_transfer_headroo
   ```

2. **Engine-enforced DR floor** inside the NonProd+DR group (Azure has no native sub-reservation): `[Decided - D7]`

   ```
   DR_Floor_vCPU(R)              = potential_dr_demand(R) × vCPU_per_instance × dr_ratio_max
   Effective_NonProd_Ceiling(R)  = NonProd_DR_Group_Limit(R) - DR_Floor_vCPU(R)
   ```

   The floor is always sized at **`dr_ratio_max` (0.40)** — the upper bound of the DR ratio range — so it never needs to grow if the operating ratio is later raised (0.30 → 0.40). Only actual Prod demand growth expands the floor. `[Decided - D7]`

3. **Hard-constraint enforcement at placement:**
   - **HC-2 CAPACITY_FLOOR** — CR headroom `2 × requested_vm_count × SKU_vCPU`.
   - **HC-3 QUOTA_FLOOR** — env-type-aware group headroom check (Prod group / effective NonProd ceiling / DR headroom). Reads from the `QuotaGroup` entity, not legacy per-subscription `QuotaRecord`. `[Decided - D9]`
   - **HC-7 DR_FLOOR_INTEGRITY** — rejects NonProd placement that would push `nonprod_quota_used` above the effective ceiling, evaluated **after** HC-3. `[Decided - D7]`

4. **Legacy per-subscription quota tracking is retained for monitoring/audit only** and superseded for all placement and scoring checks by group-level fields. `[Decided - D6]`

5. **Continuous protection:** the reconciliation loop watches for floor violations and emits `DRFloorViolationDetected`; NonProd CRG scale operations that approach the ceiling are operator-gated.

Worked POC topology (GP-06): Prod group 128 vCPU; NonProd+DR group 80 vCPU; DR floor 32 vCPU → effective NonProd ceiling 48 vCPU.

Group Accounting Formulas (normative) All four are **engine accounting controls** — they do not create a native Azure sub-reservation: `[Documented - §26]`

```
DR_Floor_vCPU             = Potential_DR_Demand × vCPU_Per_Instance × DR_Ratio_Max
Effective_NonProd_Ceiling = NonProd_DR_Group_Limit - DR_Floor_vCPU
NonProd_Headroom          = Effective_NonProd_Ceiling - NonProd_Used_vCPU
Group_Headroom            = Group_Limit - Group_Used
```

Exact Enforcement Controls

- Every **NonProd** increase performs a group check **and** a subscription check. `[Documented - §26]`
- Every **DR** increase performs a group check, subscription check, SKU check, capacity check, **and** active-incident check.
- A **separate detector** recalculates the DR floor from authoritative assignment/allocation data. Any disagreement between the command-time and detector calculations **disables automatic NonProd expansion** (fail-safe) and raises `DRFloorViolationDetected`.
- Group propagation is **polled** — no fixed propagation SLA is assumed.

`groupType` Preview Dependency (FC-11) The Quota-Group `groupType` property (`AllocationGroup` = advisory vs `EnforcedGroup` = enforced) is **preview-only** in the Azure Quota REST API and is not GA. `[Documented - Azure Quota REST API reference]` The engine's accounting controls are deliberately **engine-level and do not depend on `groupType` enforcement**. If `EnforcedGroup` is ever relied upon to drop the engine's own subscription-level check, that dependency must pass the preview-acceptance gate (POC-30) and a Decision Log entry first. Engineering constraint: pin the exact preview API version in all quota-group calls and add version-drift detection to the platform health check.

Consequences

Positive: - Prod isolation is enforced at the Azure control-plane level — a NonProd surge cannot starve Prod. - Tier 2 Emergency Capacity Transfer becomes **truly quota-neutral** (ARM operations only, no Azure approval gate) because NonProd and DR draw from the same group pool (see ADR-003). - Clean chargeback: one Azure billing record per group per region. - The DR floor guarantees a protected reservation NonProd can never erode.

Negative / trade-offs: - The DR floor is a **soft ceiling the engine must maintain continuously** — Azure won't enforce it. A bug could allow NonProd to breach the floor; mitigated by `DRFloorViolationDetected` alerting and operator gates. `[D6]` - Sizing the floor at `dr_ratio_max` over-reserves quota slightly at lower operating ratios — a deliberate reliability-over-cost trade, reviewed quarterly. `[D7]` - `Quota_Score` and HC-3 must be environment-type-aware (three code paths). - **Hard dependency on Azure Quota Groups GA** — Blocker B-1 (POC-30); if `groupQuotas` returns 404 in the target tenant/region, escalate to Azure Support.

Alternatives Considered

Alternative	Why Rejected
Single shared quota pool (all three CRGs)	Breaks Prod isolation — a NonProd surge can consume Prod's quota. `[D6]`
Three separate groups (one per CRG)	Maximum isolation but makes Tier 3 transfer non-atomic — releasing NonProd quota lands in the wrong group, forcing an Azure quota request mid-crisis (hours of RTO impact). `[D6]`
Per-subscription quota only (legacy)	Cannot make emergency transfer quota-neutral without risky cross-subscription coordination. `[D6]`
DR floor sized at current ratio	Undersizes the floor if the ratio is later increased, blocking DR expansion. `[D7]`

Appendix — Decision Log Cross-Reference

ADR	Primary Decisions	Hard Constraints	Key Gaps/Blockers
ADR-002 Quota & Capacity	D6, D7, D9	HC-2, HC-3, HC-7	B-1 (Quota Groups GA, POC-30)

Appendix — Status Legend

Status	Meaning
Proposed	Under discussion; not yet ratified
Accepted	Ratified and in force
Deprecated	No longer recommended but not yet replaced
Superseded	Replaced by a later ADR (referenced explicitly)

Appendix — Evidence Tag Taxonomy

Tag	Meaning
`[Documented]`	Traceable to Azure platform documentation or a formal FR/NFR
`[Decided]`	Explicit design choice in the Decision Log (D1–D11)
`[Derived]`	Logical consequence of a documented constraint or decision
`[Assumed]`	Architectural judgement pending POC validation

Related ADRs

- **ADR-001 — Region Selection** (`acrme_adr_001_region_selection.md`)
- **ADR-003 — Capacity Management during Disaster Recovery (DR)** (`acrme_adr_003_capacity_manag`
- **ADR-004 — Forecast and Increase of Capacity and Quota** (`acrme_adr_004_forecast_and_increase_o`

Document Status: Accepted
Next Review: After POC-30 (Quota Groups GA) and POC-31 (quota release latency), and on resolution of G-14 / G-15.